Accurate and Generalizable RNA Engineering

Deep Learning Framework for RNA Inverse Folding with Geometric Structure Potentials

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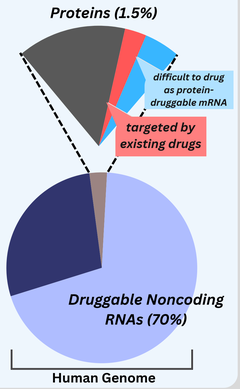
*All figures done by student.

Introduction

RNA (Ribonucleic Acid) is a highly versatile biomolecule that ***plays a fundamental role in numerous biological processes*** such as protein synthesis, gene expression regulation, catalysis, and cellular signaling.

- noncoding RNAs (ncRNAs) are a large part of the human genome
- **ncRNAs** are crucial factor in many biological functions:
  - gene regulation, chromatin organization, RNA modification
- ncRNAs ***perform functions largely through their three-dimensional (3D) structures,*** shaped by base pair interactions in their secondary structure
- Improving RNA inverse-folding will lead to enormous steps in the fields of ***vaccine creation, synthetic biology, drug design, and RNA-based therapies.***

Hence, determining which sequences can fold into a desired 3D structure is paramount to controlling an RNA's function. We call this RNA inverse folding.



Current Problems:

data scarcity

- ***vast # of possible conformations***
- ***computational methods: inaccurate + ungeneralizable***
- ***traditional methods:*** experimental trial + error (X-ray crystallography, NMR spectroscopy, and cryo-EM)
  - ***Costly + inefficient***

Purpose and Hypothesis

Problem - Traditional methods are costly and inefficient for RNA inverse folding. However, current computational methods are not accurate or generalizable enough to justify widespread use.

Research Question - *Does incorporating geometric and spatial features present in RNA enable better computationally driven RNA inverse folding?*

Hypothesis

- RNA's tertiary structure and primary structure are related
- The relationship between the two can be modelled through mathematical equations that leverage geometric and spatial features
- These features can be learned with machine learning (GVP-Transformer), which contains an architecture mirroring its function to ***predict a primary RNA sequence from a given tertiary structure***

Research Objectives

Lower costs through computational means by ***creating a new framework suited for RNA's problem structure*** that can:

1. ***satisfy specified RNA structural constraints***
2. ***generalize across different RNA families and challenges***
3. ***generate many sequences for a given structure***

Increase accuracy of RNA sequence generation by ***creating a new Machine Learning (ML) framework*** that leverages RNA's geometric and spatial features to design RNA sequences that reliably adopt target structures and model inter-nucleotide spatial relationships

Evaluate my model's performance using established metrics:

- ***Recovery Rate + TM-Score*** and different scenarios: ***Standard Benchmark + RNA-puzzles + Mask-family Validation***

Methodology

Data Processing

BGSU - RNA List Version 3.321
Download from RCSB PDB

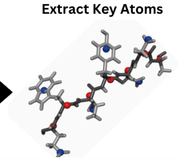


Filter lengths between 20 - 512 nucleotides

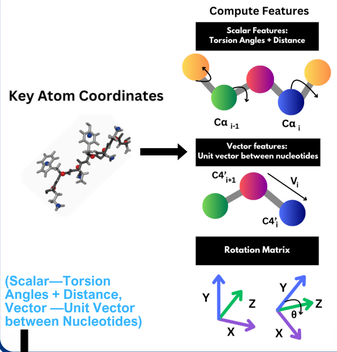
Single chain extracted from 3D structures

CD-Hit remove redundancy of 80% sequence similarity

Extract Key Atoms



Compute Features + Build Graph



Calculate the Torsion Angles:

- ϵ (epsilon): C4' → C3' → O3' → P (i+1)
- ζ (zeta): C3' → O3' → P (i+1) → O5' (i+1)
- α (alpha): O3' (i-1) → P → O5' → C5'
- β (beta): P → O5' → C5' → C4'
- γ (gamma): O5' → C5' → C4' → C3'
- δ (delta): C5' → C4' → C3' → O3'

Find the displacement vector, a 3D spatial vector, between the C4' atoms of nucleotide i and nucleotide i+1. Store in matrices s and V, where

$$s \in \mathbb{R}^{N \times d_s}$$

$$V \in \mathbb{R}^{N \times d_v \times 3}$$

Extract 3 atoms per nucleotide, N1, C4', C1', to construct a local coordinate frame
Compute the Euclidean norm (L2) along a given dimension

These vectors form the rotation matrix:

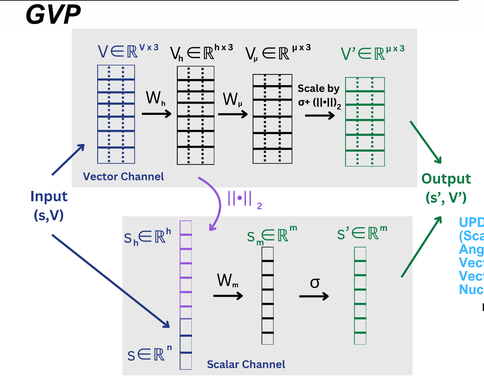
$$R = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3]$$

$\mathbf{e}_1 = \text{normalize}(\mathbf{v}_1, \text{dim} = -1)$ (unit vector along C → CA)
$\mathbf{e}_2 = \text{normalize}(\mathbf{u}_2, \text{dim} = -1)$ (adjusted vector orthogonalized against $\mathbf{e}_1$)
$\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$ (cross product ensuring a right-handed coordinate system)

Model Building and Deep Learning

Training is conducted on two NVIDIA GeForce RTX 4090 GPUs

GVP



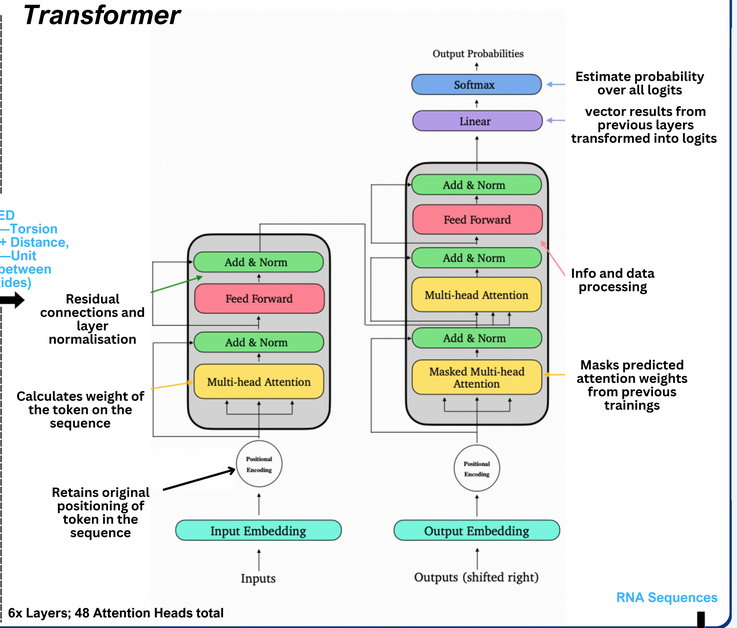
UPDATED (Scalar—Torsion Angles + Distance, Vector —Unit Vector between Nucleotides)

GVP Algorithm

$V_h \leftarrow W_h V$ $\in \mathbb{R}^{h \times 3}$
$V_\mu \leftarrow W_\mu V_h$ $\in \mathbb{R}^{\mu \times 3}$
$s_h \leftarrow \|V_h\|_2$ (row-wise) $\in \mathbb{R}^{h}$
$v_\mu \leftarrow \|V_\mu\|_2$ (row-wise) $\in \mathbb{R}^{\mu}$
$s_{h+n} \leftarrow \text{concat}(s_h, s)$ $\in \mathbb{R}^{h+n}$
$s_m \leftarrow W_m s_{h+n} + b$ $\in \mathbb{R}^{m}$
$s' \leftarrow \sigma(s_m)$ $\in \mathbb{R}^{m}$
$V' \leftarrow \sigma^+(v_\mu) \odot V_\mu$ (row-wise multiplication) $\in \mathbb{R}^{\mu \times 3}$

return (s', V')

Transformer

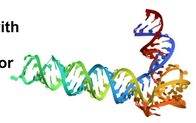


6x Layers; 48 Attention Heads total

RNA Sequences

Evaluation

Fold Back with AlphaFold3 (TM-Score for evaluation)



Initial learning Rate	1 × 10^−4
Batch Size	16
Weight decay coefficient	5 × 10^−5
Dropout	0.2
Gradient clipping global norm	1.0
Epochs	80

*early stopping triggered if the validation loss fails to improve for 5 consecutive epochs

Training and Loss

Parameters:

- ***cross-entropy*** as the loss function

$$L = -\sum_{k=1}^{K} y_k \log(p_k)$$

- Model parameters optimised using ***Adam optimizer***

Generated RNA sequences

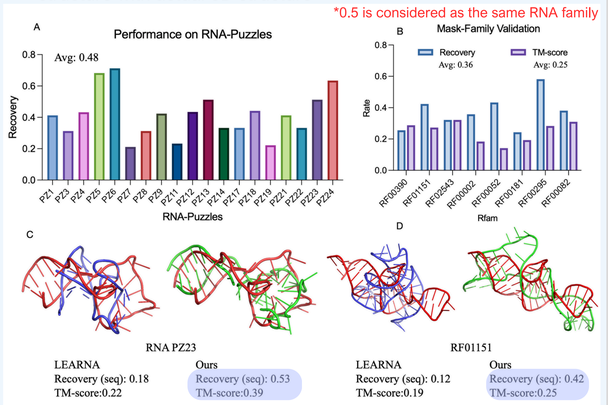


Results

Analysing how the model does in ***RNA puzzles, Standard Benchmark, and Mask-family Validation*** situations. Metrics used:*

- ***Recovery rate:*** Similarity between generated and actual sequence
- ***TM-Score:*** Similarity between generated sequence's folded 3D structure with actual 3D structure

*0.5 is considered as the same RNA family



Methods/Metrics	Recovery Rate		TM-score	
	Standard	RNA-Puzzles	Standard	RNA-Puzzles
Ours	**0.413**	**0.481**	**0.281**	**0.332**
RDesign	0.328	0.403	0.252	0.272
Ribologic	0.142	0.242	0.190	0.250
Learna	0.213	0.293	0.224	0.194

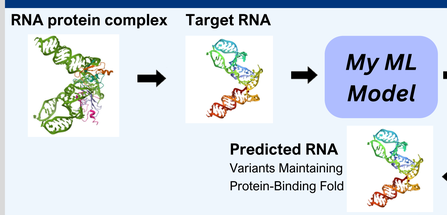
-> higher results, better accuracy, more generalizability

Conclusions

This research demonstrates that my GVP-Transformer framework can engineer RNA sequences more accurately, efficiently, and scalably than other current computational methods. My contributions:

- ***Created a novel ML model for RNA inverse folding*** that can use an RNA 3D structure to accurately generate a set of sequences that are predisposed to fold into a similar structure
  - GVP captures RNA's geometric and spatial features
  - Transformer models RNA's inter-nucleotide relationships
- ***Improved design accuracy*** from past ML models' average
  - recovery rate of 0.223 to my GVP-Transformer model's 0.413
  - TM-score of 0.238 to my GVP-Transformer model's 0.332
- ***Higher sequence generalizability*** (as shown in Mask Family Validation): my model's 0.36 compared to other ML model's 0.155

Future Work



This research has suggested possible explorations for future research:

- ***Accurate + efficient creation of RNA Vaccines***
- ***Anti-aging drug design***
- ***RNA Aptamer Design (targeted cancer therapy)***
- ***Treatment + Vaccination for Alzheimer's, Schizophrenia (genetic regulation and RNA based therapies)***